

PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947 –enacted by the Govt. of India)

SYLLABI

Course Outline for
B.Voc (SOFTWARE AND DEVELOPMENT)
as per NEP 2020

w.e.f session 2026-27

Semester-I

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
SD-101	Core Discipline	Basic Computer Skills	4	--	4	80	20	--	--	100
SD-102	Core Discipline	Logic Development Technique	2	--	4	40	10	--	--	100
SD-102P		Practical Based on SD-102	--	2		--	--	40	10	
SD-103	Core Discipline	Fundamental of Information & Technology II	--	4	4	--	--	80	20	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC I	Skill Enhancement Course	Common Basket			3					75
B.Voc MDSC 101	Multidisciplinary Course	Fundamental of Information & Technology I			3					75
VAC I	Value-Added Course	Common Basket			2					50
PW I	Project Work	Project Work-I & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-II

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
SD-201	Core Discipline	Relational Database Management System	4	--	4	80	20	--	--	100
SD-202	Core Discipline	Programming in C Language	2	--	4	40	10	--	--	100
SD-202P		Practical Based on SD-202	--	2		--	--	40	10	
SD-203	Core Discipline	RDBMS using MySQL	--	4	4	--	--	80	20	100
AEC / MIL	Ability Enhancement Course	English			2					50
SEC II	Skill Enhancement Course	Common Basket			3					75
B.Voc. MDSC 201	Multidisciplinary Course	Soft Skills and Personality Development			3					75
SIT 201*	Internship	Summer Industrial Training(SIT)*	--	4	4	--	--	80	20	100
Total Credits					**24**					**600**

Job Role ------ Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer

*Summer Industrial Training of 2-4weeks in a relevant Industry after 2nd Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 3rd Semester. Viva-Voce examination to be held within 3-weeks of the start of 3rd semester.

** Minimum Credits required to get Certificate Course in (SOFTWARE AND DEVELOPMENT)=48 + 4 #.

#Exit 1: Certificate Course-Student can exit after completion of 1st year having secured 48 Credits and will be awarded a UG Certificate provided they complete a vocational course/internship of 4 credits during the summer vacation of the first year, in addition to the 6 credits from skill enhancement courses earned during first and second semester.

***One credit of theory lecture is of 1 hour duration and one credit of practical is of 2 hours duration.

Semester-III

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
SD-301	Core Discipline	Mathematical Tools for Computer Science	4	--	4	80	20	--	--	100
SD-302	Core Discipline	Object Oriented Programming using C++	2	--	4	40	10	--	--	100
SD-302P		Practical Based on SD-302	--	2		--	--	40	10	
SD-303	Core Discipline	Data Structure using C++	--	4	4	--	--	80	20	100
AEC / MIL	Ability Enhancement Course	English/MIL			2					50
SEC III	Skill Enhancement Course	Common Basket			3					75
B.Voc. MDSC	Multidisciplinary course	Common Basket			3					75
VAC II	Value-Added Course	Common Basket			2					50
PW II	Project Work	Project Work-II & Viva-Voce	-	2	2	-	-	50	-	50
Total Credits					**24**					**600**

Semester-IV

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
SD-401	Core Discipline	Operating System	4	--	4	80	20	--	--	100
SD-402	Core Discipline	Web Programming	2	--	4	40	10	--	--	100
SD-402P		Practical Based on SD-402	--	2		--	--	40	10	
SD-403	Core Discipline	JAVA Programming	2	--	4	40	10	--	--	100
SD-403P		Practical Based on SD-403	--	2		--	--	40	10	
SD-404	Core Discipline	Computer based accounting	--	4	4	--	--	80	20	100
AEC / MIL	Ability Enhancement Course	English/MIL			2					50
VAC III	Value-Added Course	Common Basket			2					50
SIT 401	Internship	Summer Industrial Training (SIT)*	--	4	4	--	--	80	20	100
Total Credits					**24**					**600**

Job Role Technology and Information systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer

*Summer Industrial Training of 2-4 weeks in a relevant Industry after 4th Semester Examinations during summer break. Training report by the student to be submitted within one week of start of 5th Semester. Viva-Voce examination to be held within 3-weeks of the start of 5th semester.

** Minimum Credits Required to get Diploma Course in (SOFTWARE AND DEVELOPMENT) =96 + 4 #.

#Exit 2: Diploma Course-Student can exit after completion of 2nd year having secured 96 Credits and will be awarded a UG Diploma provided they complete a vocational course/internship of 4 credits during the summer vacation of the second year, in addition to the 9 credits from skill enhancement courses earned during first, second and third semester.

***One credit of theory lecture is of 1 hour duration and one credit of practical is of 2 hours duration.

Syllabus for B.Voc. (SD) – 5th and 6th Semester (NEP)

Semester-V

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
SD501	Core Discipline	Software engineering and quality assurance	2	2	4	40	10	40	10	100
SD502	Core Discipline	Computer graphics and Multimedia Applications	2	2	4	40	10	40	10	100
SD503	Core Discipline	ASP.NET using C#	2	2	4	40	10	40	10	100
SD504^	Core Discipline	Linux operating System	2	2	4	40	10	40	10	100
SD505^	Core Discipline	Critical Thinking and Elementary Statistics	2	2	4	40	10	40	10	100
SD506^	Core Discipline	Introduction to Artificial Intelligence and Data Science	4		4	-	-	100	-	100
VAC IV	Value-Added Course	VAC options			2	40	10			50

PW III	Project Work	Project Work-III& Viva-Voce			2	40	10			50
Total Credits					**24**					**600**

^Any two in 5th Semester

Semester-VI

Paper Code	Type of Course	Subject Name	Credits			Marks				
			Th	Pr	Total	Ext (Th)	Int (Th)	Ext (Pr)	Int (Pr)	Total
SD601	Core Discipline	Entrepreneurship Development Programme	2	2	4	40	10	40	10	100
SD602	Core Discipline	E- Commerce	2	2	4	40	10	40	10	100
SD603	Core Discipline	Emerging Technologies in Computing	2	2	4	40	10	40	10	100
SD604	Core Discipline	PHP Programming	2	2	4	40	10	40	10	100
SD605^	Core Discipline	Network Management	2	2	4*	40	10	40	10	100*
SD606^	Core Discipline	Computer Architecture	2	2	4*	40	10	40	10	100*
SD607^	Core Discipline	Introduction to Machine Learning	4		4*	80	20	-	-	100*
Total Credits					**24**					**600**

^Any two in 6th Semester

ESSENTIAL COMPONENTS OF COURSE OF **B.Voc**
(SOFTWARE AND DEVELOPMENT)

PROGRAMME OUTCOMES

1. Acquire various skills in the field of Information Technology. This includes skills related to various office automation tools like MS Office, PowerPoint, MS Excel, etc., and their applications.
2. Gain knowledge about the fundamentals of computer systems and their working. Further students will be able to develop programming skills with programming languages C, JAVA, Python, etc. by actually designing programs in concerned laboratories.
3. 3. Learn about computer networking, operating systems, and related areas both theoretically and with practice by performing experiments in the laboratories.
4. Design, develop, and implement data structures and algorithms for problem-solving in various domains by virtue of training in the laboratories.
5. Learn scripting and web development tools like PHP, JavaScript, etc., and apply these to design and develop real-world tools.
6. Equip themselves with the knowledge of financial accountancy and develop expertise in software like Tally.
7. Get exposure to the modern concepts of computing like Artificial Intelligence, Data Science, E-Commerce, and various Internet Technologies as part of this Programme.
8. Get industrial exposure in the field of information technology through an internship with industry.
9. Seek employment in various jobs in the Government sector as well as IT and related industries and perform various roles related to software development, testing, and maintenance.

Semester I

1	**Course Title**	**Basic Computer Skills**
2	**Course Code**	**SD-101**
3	**Typology of Course**	Core Discipline
4	**Course Duration**	60 hours of lectures for Theory Theory credits =4 (1 credit= 15 hours of theory learning based on 4hrs./week) (1 Semester=15 weeks)
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr><tr><td>80</td><td>20</td><td></td><td></td></tr></table> a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Field Technician- Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician
8	**Course Objectives**	The course aims to equip students with a profound understanding of computer fundamentals, hardware, software, and networking. To empowers the ability to analyze advantages and disadvantages, design computer system architectures, apply Input and output devices, and comprehend emerging technologies and well-rounded expertise in basics concepts.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** 1. Demonstrate a comprehensive understanding of computer fundamentals, including the historical development and characteristics of computers. 2. Apply knowledge of input and output devices, computer memory, and different types of software in practical scenarios. 3. Analyze and evaluate the functions and types of operating systems, data processing systems, and translators. 4. Demonstrate proficiency in understanding computer networks, the Internet, and emerging technologies, with the ability to classify and describe different types of networks.
10	**Pedagogy of the Course Work**	Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT– I

Computers: Characteristics, Historical development of computers, Evolution of computing devices, Generations of Computers: First Generation (Vacuum Tubes, Batch Processing), Second Generation (Transistors), Third Generation (Integrated Circuits, Time-Sharing Systems), Fourth Generation (Microprocessors, Personal computers), Fifth Generation (Artificial intelligence). Hardware, Software and Firmware, Data and Information. Computer Codes: Decimal System, Binary System, Hexadecimal System, Octal System and ASCII Code. Decimal to Binary conversion, Binary to Decimal Conversion.

UNIT– II

Introduction to Computer Networks: Definition and Importance of Computer Networks, Basic Components of a Computer Network, Classification of Networks, Network Topologies. Internet: Historical Overview of The Internet, Evolution of The World Wide Web (WWW), WWW Components, Internet Governance and Standards, Internet Services and Applications (Email Protocols, Telnet, FTP, IRC, Usenet News). Internet Access Methods (Dial-Up, DSL, Cable, Fiber-Optic), Browser Technologies and Features.

UNIT– III

Computer Virus: Types of Viruses, Impact of Virus, Virus Detection and Prevention techniques, Malware. Antivirus and Its Types. Multimedia: Hardware and Software for Multimedia, Multimedia: Component of Multimedia, Basic File Formats, Audio File Formats, Image File Formats, Video File Formats. Multimedia Storage Devices.

UNIT-IV

Emerging Technologies: Basics of Internet of Things (IoT), Components of IoT, Applications of IoT, Basics of Parallel Computing and Distributed Computing, Difference between Parallel computing and Distributed Computing, Advantages and Limitations of Parallel computing and Distributed Computing, Cloud Computing: Component of Cloud Computing, Types of Cloud Computing, and their applications.

13 Suggested Resources for Reading

1. Fundamentals of Computers, V. Rajaraman, PHI Publication.
2. Computer Fundamentals, P. K. Sinha, BPB Publication.
3. Computers Today, Basandra, S. K., Galgotia Publications.
4. Fundamentals of Information Technology, Anshuman Sharma, Lakhanpal Publications.
5. PC Hardware: The Complete Reference, Craig Zacker., TMH Publication.

Note: Latest edition of text books may be used.

1	**Course Title**	**Logic Development Technique**
2	**Course Code**	**SD-102**
3	**Typology of Course**	Core Discipline
4	**Course Duration**	30 hours of lectures for Theory + 60 hours of practicals Theory credits =2 and Practical credits=2 (1 credit = 15 hours of theory learning based on 2 hrs./week) + (1 credit = 30 hours of practical based on 4 hrs./week) (1 Semester = 15 weeks) 2-groups will be created for the purpose of Practical for each unit
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr><tr><td>40</td><td>10</td><td></td><td></td></tr></table> a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer
8	**Course Objectives**	This course aims to provide comprehensive knowledge of computer peripherals, hardware components, and system management. Hardware Fundamentals enable to learn about hardware technologies, their implementation/installation, uses, troubleshooting and maintenance.
9	**Course Outcomes**	On successful completion of the course, the students will be able to 1. To develop the logic of the different problems. 2. To equip students with the basic knowledge of logic development techniques. 3. To enhance the skills of students for software development.
10	**Pedagogy of the Course Work**	Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.
11	**Instructions for paper setters and candidates**	4. The syllabus of this paper has been divided into FOUR units. 5. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 6. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 7. All questions carry equal marks.
12	**Course Content**	**UNIT-I** **Logic Development Tools:** Algorithm Development: Types of Algorithms, Advantage and Disadvantage of Algorithm, Analysis of Algorithms, Complexity of Algorithm, Big-O Notation. **Flowchart:** Types of Flowcharts, Advantage and Disadvantage of Flowchart, Designing Pseudo code, Decision table, Steps for problem solving, Stepwise refinement, Modular Programming: top down & bottom-up approach. **UNIT-II** **System Analysis and Design:** Definition of problem, Feasibility study, Requirement analysis, Designing, Coding, Testing, and Implementation of software project.

UNIT-III

Basics of 'C' Programming Language Usage of Character Set, Meaning of Keywords and Identifiers, Role of Data Types, Constants and Variables. Importance of type Casting. Different Types of Operators and their Precedence, Expressions, Conditional Statements

UNIT-IV

Control Statements: Decision Making statements (if, if-else, else if, nested if), Looping Statements (For, While, do-while), Usage of Exit, Continue, Break and Goto Statement. Introduction of array and functions

13 Suggested Resources for Reading

1. Kanetkar, Y.P., *Let us C*, BPB Publications
2. Balagurusami, E., *Programming in ANSI C*, Fourth Edition, Tata McGraw Hill
3. Sharma, A., *Learn Programming in C*, Lakhanpal Publisher
4. Kernighan, B.W., & Richie, D.M., *The C Programming Language*, PHI Publication
5. R. S. Salaria, *Problem Solving and Programming in C*, Khanna Publishing Pvt. Ltd.
6. Awad, E. M., *System Analysis and Design* , Galgotia Publication Pvt. Ltd.

Note: Latest edition of text books may be used.

SD-102P (Practical Based on SD-102)

Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation.

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three questions.
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

<table>
<tr><th>1</th><th>Course Title</th><th>Fundamentals of Information Technology II</th></tr>
<tr><td>2</td><td>Course Code</td><td>SD-103</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration</td><td>120 hours of practicals
Practical credits=4
(1 credit = 30 hours of practical based on 8 hrs./week)
(1Semester=15weeks)
2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td><table><tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr><tr><td></td><td></td><td>80</td><td>20</td></tr></table>
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>7</td><td>Job Role</td><td>Field Technician- Computing & Peripherals</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>The course aims to provide a comprehensive understanding of computer systems, covering fundamental hardware components. It develops practical skills in connecting peripherals, analyze and solve hardware issues logically, and assemble, configure, troubleshoot, and maintain functional computer systems. Additionally, the course focuses on installing Windows, managing drivers, and implementing antivirus software for system security.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
1. Define and explain the fundamental components of a computer system, demonstrating a holistic understanding of hardware architecture.
2. Demonstrate practical skills in connecting and configuring peripherals, including mouse, keyboard, display devices, scanning devices, and hard disk.
3. Develop critical thinking skills to analyze and solve hardware-related problems, applying a logical and systematic approach to diagnose and address issues.
4. Assemble, configure, troubleshoot, and maintain a functional computer system, integrating knowledge from all units for practical application.
5. Successfully install the Windows operating system, manage the installation and updating of drivers, install and update anti-virus software.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content

UNIT-I

Operating Systems- Windows and Unix. Fundamentals of windows, anatomy of windows, Operations on window: Opening, Minimizing, Maximizing, Moving, Resizing, closing; Windows Explorer, Folders: Creating, deleting, copying, renaming folders, folder properties; Icons, Menu, Taskbar, Control panel, Recycle bin. Overview of UNIX structure, general purpose UNIX commands such as date, echo, cal, bc, pwd, passwd; file and directory commands such as ls, mkdir, cp, mv, rm. process management commands such as ps, kill, nohup, communication
commands such as news, mesg, wall; working with editor introduction to shell programming

UNIT-II

Word Processing Package: Basics of Word Processing; Word Processing Basics; Text creation, Manipulation, Finding and replacing, Formatting of text; Printing of word document, PageLayout: Margin setting, Alignments, Adding Borders and shading, Adding Headers and Footers, setting up Multiple columns, working with tables, Spell check, Auto Correct, Grammar facility, Retrieving often used text; Auto text character formatting, language setting and thesaurus;Macros; Mail merge.

UNIT-III

Spreadsheet Package: Worksheet Basics, Data Entry in Cells: Entry of numbers, text and formulae, moving data in a worksheet, Moving around in a worksheet, Selecting Data Range. Using the interface (Toolbars, Menus), Editing Basics, Working with workbooks, Cell referencing: Absolute, Relative and Mixed; Formatting and Calculations: using Autofill, Working with Formulae, Efficient Data Display with Data formatting (number formatting, date formatting etc.), Function: different types of functions and their usage Worksheet Printing. Working with Graphs and Charts: Creating Chart and graphs, using chart wizard, sizing and moving parts, updating charts, Changing chart types, Formatting Charts. Database Management; Finding records with Data form, Adding/Deleting Records, Filtering Records in a worksheet, Sorting; Macros: Creating Macros, Record Macros, Running Macros, Assigning Macros to Buttons.

UNIT-IV

Presentation Packages: Basics, General Features, Creating a presentation, Different types of slide views, Master Slides and its use, Formatting Slides: slide design, Layout and background; Animation effect, Transition effect, timing effects, Macros.

Database Package: Introduction to Database, Tables, Data Types, Attributes, Records; Overview of MS-ACCESS, Creating Database, Creating Tables, Data types, Importing andExporting data, using Wizards, Creating forms and queries.

13 Suggested Resources for Reading

1. Sinha, P.K., *Computer Fundamentals*, BPB Publications
2. Rajaraman, V., *Fundamentals of Computers*, PHI
3. Mathur, R., *DOS 6.2 Quick Reference*, Galgotia Publicaitons Pvt. Ltd.
4. Sharma, A., *Fundamentals of Information Technology*, Lakhanpal Publisher
5. Texali, R.K.: *PC Software Made Simple*, Tata McGraw Hill.

Note: Latest edition of text books may be used.

Semester II

<table>
<tr><th>1</th><th>Course Title</th><th>Relational Database Management System</th></tr>
<tr><td>2</td><td>Course Code</td><td>SD-201</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration</td><td>60 hours of lectures for Theory
Theory credits =4
(1 credit= 15 hours of theory learning based on 4hrs./week)
(1 Semester=15 weeks)</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>100Marks (80TH+20IA)
<table><tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr><tr><td>80</td><td>20</td><td></td><td></td></tr></table>
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>7</td><td>Job Role</td><td>Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>1. To provide good understanding of database design, modelling and implementation of database management systems.
2. To educate students with the fundamental concepts of Data Base Design, Data Models and different Data Base Languages.
3. To make access to the data easy for the user.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
1. Understand the basic concepts of database management systems.
2. Apply SQL to find solutions to a broad range of queries.
3. Apply normalization techniques to improve database design.
4. Analyze a given database application scenario to use ER model for conceptual design of the database.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Classroom Lecture, Interactive sessions, PPT, Group Discussions etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content

UNIT-I

Introduction: Database vs. File Oriented Approach, Advantages and Disadvantages of DBMS, Data models: Hierarchical, Network and Relational models.

UNIT - II

Entity Relationship model, Client-Server Architecture, Data Independence. Concept of Relational Algebra and Calculus, Storage Organization for Relations

UNIT-III

Relational Algebra: Operations - union, intersection, difference, Cartesian product, projection, selection, division and relational algebra queries; Relational Calculus: Tuple oriented and domain oriented relational calculus and its operations, Normalization and its forms. DBMS vs RDBMS

UNIT-IV

Database Integrity, Security, Concurrency, Backup and Recovery of the database. Design and structure of Distributed databases

13 Suggested Resources for Reading

1. Desia, B.C., *An Introduction to Database System*, Galgotia Publications ,
2. Date, C.J. & Sharma, A., *Database System Vol I & II,* Narosa Publications
3. Sharma, A., *Fundamentals of DBMS,* Lakhanpal Publisher
4. Korth, H.F., *Database System Concepts*, McGraw Hill,

Note: Latest edition of text books may be used.

<table>
<tr><td>1</td><td>Course Title</td><td>Programming in C Language</td></tr>
<tr><td>2</td><td>Course Code</td><td>SD-202</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration</td><td>30 hours of lectures for Theory + 60 hours of practicals
Theory credits =2 and Practical credits=2
(1 credit = 15 hours of theory learning based on 2 hrs./week) +
(1 credit = 30 hours of practical based on 4 hrs./week)
(1 Semester = 15 weeks)
2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td><table><tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr><tr><td>40</td><td>10</td><td></td><td></td></tr></table>
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>7</td><td>Job Role</td><td>Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>1. To help the students in finding solutions to the various real life problems and converting the solutions into computer program using C.
2. To acquire knowledge about the basic concept of writing a program.
3. To learn and acquire the art of computer programming.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to

1. Develop a C program
2. Control the sequence of the program and give logical outputs
3. Implement strings in your C program
4. Store different data types in the same memory
5. Manage I/O operations in your C program
6. Repeat the sequence of instructions and points for a memory location
7. Apply code reusability with functions and pointers
8. Understand the basics of file handling mechanisms
9. Explain the uses of pre-processors and various memory models</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Class Room Lecture, Laboratory, Interactive sessions, PPT, Practical and Group Discussions etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content

UNIT-I

Fundamental of 'C': Problem Solving steps, Structure of C Program, **I**/O statements, assignment statements, C character Set, Constants, Variables, Operators and Expressions, Standards and Formatted statements, Keywords, Data Types and Identifiers.

Control Structures: Introduction, Decision making with if – statement, if-else and Nested if, while and do-while, for loop. Jump statements: break, continue, goto, switch Statement.

UNIT-II

Functions: Function Declaration and Definition, types of functions, Call by value and Call by reference , Recursion, Pre-processor Directives.

UNIT-III

Preprocessor Directives: Introduction and Use, Macros, Conditional Preprocessors and Header Files.
Arrays: Introduction, Types, Operations on Arrays.

UNIT-IV

Structure and Union: Declaring structure, accessing structure members, Array of structure, nested structure, Unions.

Files: introduction, creating, opening of data files, closing a data file and file handling functions.

13 Suggested Resources for Reading

1. Kanetkar, Y.P., *Let us C*, BPB Publications
2. Balagurusami, E., *Programming in ANSI C*, Fourth Edition, Tata McGraw Hill
3. Gottfried, B.S., *Programming in C*, McGraw Hills.
4. Sharma, A., *Learn Programming in C*, Lakhanpal Publisher
5. R. S. Salaria, *Problem Solving and Programming in C*, , Second Edition

Note: Latest edition of text books may be used.

SD-202P (Practical Based on SD-202)

Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation.

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three questions.
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

<table>
<tr><th>1</th><th>Course Title</th><th>RDBMS using MySQL</th></tr>
<tr><td>2</td><td>Course Code</td><td>SD-203</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration</td><td>120 hours of practicals
Practical credits=4
(1 credit = 30 hours of practical based on 8 hrs./week)
(1Semester=15weeks)
2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td>
<table>
<tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr>
<tr><td></td><td></td><td>80</td><td>20</td></tr>
</table>
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the Examination Hall is allowed. The calculators will not be provided by the University/College to the examinees
</td></tr>
<tr><td>7</td><td>Job Role</td><td>Data Entry operator, IT Help Desk, Office Executive, Software Trainee, Technical Support Voice, B.P.O, Lab Technician, Database Administrator, Network Administrator, Application Programmer</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>1. To introduce principles and foundations of databases, including architecture, design issues, integrity control, query processing and optimization, transactions, and concurrency control.
2. To design and implement database for enterprise application.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
1. Understanding Database Concepts
2. Understanding Database Storage & Database Security
3. Understanding Entities and Relationships & Normalization
4. Understanding the Relational Data Model
5. Understanding Database Design and Performance Tuning
6. Understanding Database Backup and Restore & MySQL</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content

UNIT-I

Database Fundamentals: What is a database? What is MySQL?, Introduction to MySQL & MySQL workbench.
Database Design: Introduction To Database Design, Learn Database Normalization with the help of a case study, Learn ER Modelling with a Case Study

UNIT-II

MySQL Basics: Creating a Database & MySQL Data Types, SELECT in MySQL, How to query data using Where clause in MySQL, Insert Into in MySQL, Delete & Update data in MySQL
Data Sorting: How sorting is done in MySQL, using ORDER BY, DESC and ASC, Group By in MySQL, Wildcards in MySQL, Using Regular Expressions & Wild Cards in MySQL

UNIT-III

Functions:, Introduction to functions, Aggregate Function inMySQL, Null value & Keyword in MySQL, Auto Increment in MySQL, Alter, Drop & Rename function in MySQL, Limit keyword in MySQL

Unit-IV

Queries: Using Sub-Queries in MySQL, Joins in MySQL, Unions inMySQL, Views in MySQL, Index in MySQL **Minor Application Design:** Creating an Application using MySQL

13 Suggested Resources for Reading

1. Desai, B.C., *An Introduction to Database System*, Galgotia Publications.
2. Korth, H.F., Database System Concepts, McGraw Hill
3. Date, C.J., Kannan, A. & Swamynathan, S., *An Introduction to Database system*, Pearson Publications
4. Jain, M., Pillai, V., Singh, S. & Jain,S., *Introduction to Database Management*, BPB.
5. Schwartz, B., & Zaitsev, P., *High Performance MySQL: Optimization, Backups, Replication, and More*.
6. Dyer, R.J.T., *MySQL in a Nutshell*, O'Reilly.

Note: Latest edition of text books may be used.

SD-202P (Practical Based on SD-202)

Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation.

5. Examiner will set total of four questions covering the whole syllabus.
6. Student will attempt any three questions.
7. Each question carries 5 marks.
8. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

Semester- III

1	**Course Title**	Mathematical Tools for Computer Science
2	**Course Code**	**SD-301**
3	**Typology of Course**	Core Discipline
4	**Course Duration**	120 hours of practicals Practical credits=4 (1 credit = 30 hours of practical based on 8 hrs./week) (1Semester=15weeks) 2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr><tr><td>80</td><td>20</td><td></td><td></td></tr></table> a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the ExaminationHall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Information and system operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer.
8	**Course Objectives**	1. To develop arithmetic, algebraic, geometric, and problem-solving skills. 2. To apply problem solving and logical skills. 3. To make the students familiar with several subfields of mathematics (e.g. numerical analysis, topology, operations research).
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** • identify the logical form of statements in propositional and predicate logic • write proofs, including inductive proofs, of properties of numbers, sets, functions, relations, and graphs • understand the steps of some cryptographic algorithms • perform elementary matrix operations.
10	**Pedagogy of the Course Work**	Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

Unit I

Sets and Relations:- Definition of sets, subsets, complement of a set, universal set, intersection and union of sets, De-Morgan's laws, Cartesian products, Equivalent sets, Countable and uncountable sets, minset, Partitions of sets. Relations: Basic definitions, graphs of relations, properties of relations.

Unit II

Matrix:- Introduction, Different kinds, matrix addition and scalar multiplication, multiplication of matrices, transpose etc. Square matrices, inverse and rank of a square matrix, differentiation and Integration.

Unit III

Numerical Methods: Solving simultaneous equations using Gauss elimination, Gauss Jordan Methods, Matrix Inversion method.

Unit IV

Linear Programming:- Graphical and Simplex method, Duality in Linear programming, Transportation problems, Assignment and travelling salesman problem.

13 Suggested Resources for Reading

1. Hiller, F.S & Liberman,G.J: Introduction To Operations Research, 2nd Day Inc London. Edition Holden
2. Schaum Serierds: Theory and Problems of Essential Computer Mathematics, Mc Graw Hill, New York, 3 Edition.
3. Salaria, R.S: Computer Oriented Numerical Methods, Khanna Book Publishing Co (P) Ltd, New Delhi.
4. Liu, C. L.: Elements of Discrete Mathematics, McGraw Hill.

Note: Latest edition of text books may be used.

1	**Course Title**	Object Oriented Programming using C++
2	**Course Code**	**SD-302**
3	**Typology of Course**	Core Discipline
4	**Course Duration**	120 hours of practicals Practical credits=4 (1 credit = 30 hours of practical based on 8 hrs./week) (1Semester=15weeks) 2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr><tr><td>40</td><td>10</td><td></td><td></td></tr></table> a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the ExaminationHall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Information and systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer.
8	**Course Objectives**	1. To help the students gain a better understanding of OO design and program implementation by using OO language features. 2. To make students learn various concepts of object oriented approach towards problem solving. 3. To introduce different techniques pertaining problem solving skills.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** • Describe OOPs concepts • Use functions and pointers in your C++ program • Understand tokens, expressions, and control structures • Explain arrays and strings and create programs using them • Describe and use constructors and destructors • Understand and employ file management • Demonstrate how to control errors with exception handling
10	**Pedagogy of the Course Work**	Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

Unit I

Concepts of Object Oriented Programming: Introduction to OOP, Difference between OOP and Procedure Oriented Programming, Object, Class, Encapsulation, Data Hiding, Abstraction, Polymorphism, Inheritance. Analysis and design of system using Object Oriented Approach, Structure of a C++ Program and I/O streams.

Unit II

Classes: Class declaration: Data Members, Member Functions, Private and Public Members, Data Hiding and Encapsulation, Arrays within a class. Objects: Creating objects, Accessing class data members, Accessing member functions, Array of objects, Objects as function arguments: Pass by value, Pass by reference, Pointers to Objects.

Constructors and Destructors: Constructors: Declaration and Definition, Default Constructors, Parameterized Constructor, Copy Constructor and Destructors.

Unit III

Polymorphism: Classification, static and dynamic polymorphism, virtual functions, pure virtual functions. Function Overloading and Operator Overloading: Function Overloading: Declaration and definition. Operator Overloading: Overloading unary, binary, arithmetic, relational operators, Operator Overloading using friend function.

Unit-IV

Inheritance: Extending classes: Concept of Inheritance, base class, derived class, defining derived classes, visibility modes, private, public and protected

13 Suggested Resources for Reading

1. Anshuman Sharma : Learn Programming in C++, Lakhanpal Publisher.
2. E. Balaguruswamy : Object Oriented Programming with C++, TMH.
3. Bjarne Stroustrup : The C++ Programming Language, Addison Wesley.
4. Robert Lafore : OOP in Turbo C++,Galgotia.
5. Herbert Schildt : C++ The Complete Reference, TMH

Note: Latest edition of text books may be used.

SD-302P (Practical Based on SD-302)

Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book and Assignments in evaluation.

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three questions.
3. Each question carries 5 marks.
4. Practical file carries 5 marks and 20 marks for practical work and viva-voce.

1	**Course Title**	Data structure using C++
2	**Course Code**	**SD-303**
3	**Typology of Course**	Core Discipline
4	**Course Duration**	120 hours of practicals Practical credits=4 (1 credit = 30 hours of practical based on 8 hrs./week) (1Semester=15weeks) 2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr><tr><td></td><td></td><td>80</td><td>20</td></tr></table> a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the ExaminationHall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Information and systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer.
8	**Course Objectives**	1. To teach students various data structures and explain to them algorithms for performing various operations on these data structures. 2. To use various data structures effectively in application programs. 3. To determine which algorithm or data structure to be used in different scenarios.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** • Able to identify the appropriate data structures and algorithms for solving real world problems. • Able to implement various kinds of searching and sorting techniques. • Able to implement data structures such as stacks, queues, Search trees, and hash tables to solve various computing problems.
10	**Pedagogy of the Course Work**	Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

Unit I

Introduction of Data Structure: Definition of Data Structure, types of Data Structure. Operations on data structure. Arrays: Linear and multi-dimensional arrays and their representation, operations on arrays. Sorting and searching in array.

Unit II

Stacks & Queues: Definition of stacks Sequential and linked representations, operations on stacks, Queues: Sequential representation of queue, linear queue, circular queue, operations on linear and circular queue, linked representation of a queue and operations on it, priority queue. Linked List: Linear linked list, operations on linear linked list, doubly linked list, operations on doubly linked list.

Unit III

Trees: Inserting a node, deleting a node, AVL trees, Heap Trees Practical implementation.

Unit IV

Graphs: Traversal of a graph (breadth-first search and depth-first search), and applications of graphs.

13 Suggested Resources for Reading

1. Lipschuitz L. Seymour: Data Structure, Schaum Outline Series, TMH, New Delhi.
2. Sharma Anshuman : Data Structures Using C++, Lakhanpal Publishers.
3. Salaria, R. S: Test Your Skills in Data Structures; Khanna Book Publishing Co. (P.) Ltd., New Delhi.
4. Sofat Sanjeev: Data Structure with C and C++, Khanna Book Publishing Co

Note: Latest edition of text books may be used.

1	**Course Title**	Operating System
2	**Course Code**	**SD-401**
3	**Typology of Course**	Core Discipline
4	**Course Duration**	120 hours of practicals Practical credits=4 (1 credit = 30 hours of practical based on 8 hrs./week) (1Semester=15weeks) 2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr><tr><td>80</td><td>20</td><td></td><td></td></tr></table> a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the ExaminationHall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Information and systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer.
8	**Course Objectives**	1. To introduce students with the basic concepts of Operating System, its functions and services. 2. To make students understand the underlying principles, techniques and approaches which constitute a coherent body of knowledge in operating systems. 3. To make students understand how operating system is used as a resource manager.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** • Learn concepts of operating systems • Learn the mechanisms of OS to handle processes • Study of various mechanisms involved in memory management techniques • Gaining knowledge of deadlocks prevention and detection techniques • Analyzing disk management functions and techniques
10	**Pedagogy of the Course Work**	Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.
11	**Instructions for paper setters and candidates**	1. The syllabus of this paper has been divided into FOUR units. 2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. 4. All questions carry equal marks.

12 Course Content

UNIT I

Introduction to Operating System:- OS, History & Types of OS, Functions/Operations of OS, Users services/jobs, Segmentation, Virtual memory and Page Replacement algorithms, Scheduling:-Process states, virtual processors, interrupt mechanism, scheduling algorithms- Pre-emptive and Non pre- emptive scheduling; Scheduling Algorithms: FCFS, SJFS, Priority scheduling, Multilevel queue scheduling, Multilevel feedback queue scheduling.

UNIT II

System Deadlock:-System Model, Deadlock Characterization: Necessary conditions, Resource Allocation graph, Method for handling deadlock, Deadlock prevention: Mutual Exclusion, Hold and wait, No Pre-emption, Circular wait, Deadlock Avoidance: Safe state, Resource Allocation graph Algorithm, Banker's Algorithm; Deadlock Detection, Recovery from deadlock.

UNIT III

Memory Management: Hierarchy of memory types, Cache memory: Types: Associative memory, direct mapped, set associative. Memory Allocation: Address binding, Address Space, Memory Protection, Contiguous and Non-Contiguous allocation, Swapping, Fragmentation; Paging: Protection, Shared pages, Techniques for structuring of page table.

UNIT IV

Segmentation: Segmentation with paging; Virtual Memory: Demand paging; Page replacement Algorithms: FIFO, Optimal, LRU, LFU, MFU, Working set, Thrashing

13 Suggested Resources for Reading

1. Silberschatz and Peter B.Galvin, "Operating System Concepts" Addison Wesley Publishing Company

2. Dhamdhere, —Systems Programming & Operating Systems" Tata McGraw Hil

3. Sharma Anshuman -Fundamentals of Operating System, Lakhanpal Publisher

4. Operating System by Madnick Donovan.

Note: Latest edition of text books may be used.

<table>
<tr><td>1</td><td>Course Title</td><td>Web Programming</td></tr>
<tr><td>2</td><td>Course Code</td><td>SD-402</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration</td><td>120 hours of practicals
Practical credits=4
(1 credit = 30 hours of practical based on 8 hrs./week)
(1Semester=15weeks)
2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td><table><tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr><tr><td>40</td><td>10</td><td></td><td></td></tr></table>
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the ExaminationHall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>7</td><td>Job Role</td><td>Information and systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer.</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>1. To introduce the whole range of web technologies starting from HTML, DHTML, JAVA SCRIPT.
2. To focus on practical aspects of the web technologies.
3. To make the students understand the various steps in designing a creative and dynamic website.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to

• Develop web pages using HTML, DHTML and Cascading Styles Sheets
• Develop a dynamic web pages using JavaScript (client side programming)
• Build and consume web services.</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content

SECTION – A

Basic terminology: About Web Server; Web Client/Browser, Understanding how a Browser communicates with a Web Server, Website, Webpage, Static Website, Dynamic Website, Internet, Intranet, Extranet, WWW, URL

HTML: Structure of an HTML program, Paragraph Breaks, Line Breaks; Emphasizing Material in a Web Page (Heading Styles, Drawing Lines); Text Styles (Bold, Italics, Underline); Other Text Effects (Centering (Text, Images etc.) Lists: Unordered List, Ordered Lists, Definition lists Adding Graphics to HTML Documents using the Border, Width, Height, Align, ALT Attributes Tables: Caption Tag, Width, Border, Cell padding, Cell spacing, BGCOLOR, COLSPAN and ROWSPAN Attributes.

SECTION – B

Linking Documents: Anchor tag, External Document References Resources:, Internal Document References Resources: and Image Maps Frames: Introduction to Frames: The <FRAMESET> tag, The <FRAME> tag, Targeting Named Frames DHTML: Introduction to cascading style sheets (CSS), Style tag, Link tag, Types of CSS: In-Line, Internal, External Forms: Attributes of Form element, Input element, The Text Element, Password, Button, Submit Button, Reset Button, The Checkbox, Radio, TextArea, Select and Option

SECTION – C

Java Script: Introduction and Features of JavaScript, Writing JavaScript into HTML, tokens, data types, variables, operations, control constructs, strings arrays, functions, core language objects, client side objects, event handling. Applications related to client side form validation.Other Built-In Objects in JavaScript: The String Object, The Math Object, The Date Object;

SECTION – D

Introduction to Dreamweaver: Understanding Workspace Layout, Managing Websites, Creating a Website, Using Dreamweaver Templates, Adding New WebPages, Text and Page Format, Inserting Tables, Lists, Images, Adding Links. Web Hosting: Understanding Domain Name & Web Space, Getting a Domain Name & Web Space (Purchase or Free), Uploading the Website to Remote Server, Introduction to Open Source Third party FTP Tools

13 Suggested Resources for Reading

1 Wanger & Wyke : Java Script Unleased, Pearson Education, New Delhi.
2 Schildt , Herbert : The Complete Reference Java 2, TMH, Latest reprint
3 Thomas Powell : HTML & CSS: The Complete Reference
4 John Pollock : JavaScript, A Beginner's Guide Dreamweaver CS5 For Dummies Paperback
5 Janine C. Warner : Edition Adobe Dreamweaver CS5 Bible Paperback
6 Joseph Lowery : Edition
7 David Powers : The Essential Guide to Dreamweaver CS4

Note: Latest edition of text books may be used.

SD-402P (Practical Based on SD-402)

Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book andAssignments in evaluation.

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three questions.
3. Each question carries 5 marks.

Practical file carries 5 marks and 20 marks for practical work and viva-voce.

<table>
<tr><td>1</td><td>Course Title</td><td>Java Programming</td></tr>
<tr><td>2</td><td>Course Code</td><td>SD-403</td></tr>
<tr><td>3</td><td>Typology of Course</td><td>Core Discipline</td></tr>
<tr><td>4</td><td>Course Duration</td><td>120 hours of practicals
Practical credits=4
(1 credit = 30 hours of practical based on 8 hrs./week)
(1Semester=15weeks)
2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit</td></tr>
<tr><td>5</td><td>Credits</td><td>4 Credits</td></tr>
<tr><td>6</td><td>Max. Marks and Time</td><td><table><tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr><tr><td>40</td><td>10</td><td></td><td></td></tr></table>
a. The exam of this course will be of 3hours duration.
b. There will be no objective type questions.
c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination.
d. Use of non-programmable calculators by the students in the ExaminationHall is allowed. The calculators will not be provided by the University/College to the examinees</td></tr>
<tr><td>7</td><td>Job Role</td><td>Information and systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer.</td></tr>
<tr><td>8</td><td>Course Objectives</td><td>1. To write programs in Java using object-oriented paradigm.
2. To familiarize students with the fundamentals of Java programming, program design and problem-solving.</td></tr>
<tr><td>9</td><td>Course Outcomes</td><td>On successful completion of the course, the students will be able to
• Able to solve real world problems using OOP techniques.
• Able to understand the use of abstract classes.
• Able to solve problems using java collection framework and I/o classes.
• Able to develop multithreaded applications with synchronization.
• Able to develop applets for web applications.
• Able to design GUI based applications</td></tr>
<tr><td>10</td><td>Pedagogy of the Course Work</td><td>Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.</td></tr>
<tr><td>11</td><td>Instructions for paper setters and candidates</td><td>1 The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.</td></tr>
</table>

12 Course Content

Unit I

Introduction to java: Definition, Comparison of java with C++, byte code, java virtual machine, constants ,variables, data types, operators, expressions, control structures, concept of class, creating objects, accessing class members, constructors, method overloading.

Unit II

Packages and interfaces: Defining a package, understanding CLASSPATH, access protection: importing packages, interfaces, defining an interface, implementing interfaces, applying interfaces, variables in interfaces. Exception handling: Definitions, types, try and catch multiple try and catch, built in exceptions.

Unit III

Multithreading: Java thread model, thread priorities, synchronizations, messaging, thread class and runnable interface, main thread: creating a thread, implementing runnable extending thread, creating multiple threads.

Unit IV

I/O applets: I/O Basics: streams, predefined streams, reading console I/P, writing console O/P, print writer class, reading and writing files, applet fundamentals, AWT controls, layout managers and menus, string handling and event handling.

13 Suggested Resources for Reading

1. Phillips LEE and Darmell Rick: Computer Graphics, Second Edition, PHI, New Delhi.
2. Daniel Dang,: An Introduction to Java Programming, PHI, New Delhi.
3. Anshuman Sharma:Learn Programming in JAVA, Lakhanpal Publisher
4. Balaguruswamy, E.: Programming with Java, A Primer, TMH, New Delhi
Wanger & Wyke,: Java Script Unleased, Techmedia, New Delhi.

Note: Latest edition of text books may be used.

SD-403P (Practical Based on SD-403)

Note: Paper will be set at the time of examination. Due weight-age may be given to the Practical note-book andAssignments in evaluation.

1. Examiner will set total of four questions covering the whole syllabus.
2. Student will attempt any three questions.
3. Each question carries 5 marks.

Practical file carries 5 marks and 20 marks for practical work and viva-voce.

1	**Course Title**	COMPUTER BASED ACCOUNTING
2	**Course Code**	**SD-404**
3	**Typology of Course**	Core Discipline
4	**Course Duration**	120 hours of practicals Practical credits=4 (1 credit = 30 hours of practical based on 8 hrs./week) (1Semester=15weeks) 2-groups will be created for the purpose of Practical for each unit 2-groups will be created for the purpose of Practical for each unit
5	**Credits**	4 Credits
6	**Max. Marks and Time**	<table><tr><td>External Theory</td><td>Internal Theory</td><td>External Practical</td><td>Internal Practical</td></tr><tr><td></td><td></td><td>80</td><td>20</td></tr></table> a. The exam of this course will be of 3hours duration. b. There will be no objective type questions. c. Students are required to have the knowledge of the developments in the subject up to 6 months before the examination. d. Use of non-programmable calculators by the students in the ExaminationHall is allowed. The calculators will not be provided by the University/College to the examinees
7	**Job Role**	Information and systems operation manager, Software Programmer, System Analyst, I.T. Officers, Computer Assistants for the General Office & Accounts Management, Database Administrator, Network Administrator, Website Development Programmer.
8	**Course Objectives**	1. To equip the students with the basic knowledge of accounting and implement it by using software package TALLY.
9	**Course Outcomes**	**On successful completion of the course, the students will be able to** • Equip students in use of accounting software for general accounting requirements. • Enable students handle manufacturing concern accounting, service accounting and so on. • Students will be equipped with the basics of the accounting cycle through accounting software.
10	**Pedagogy of the Course Work**	Laboratory, Interactive sessions, PPT, Practical and Group Discussions, etc.
11	**Instructions for paper setters and candidates**	5. The syllabus of this paper has been divided into FOUR units. 6. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus. 7. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1. All questions carry equal marks.

12 Course Content

UNIT I

Accounting: concepts and conventions, double entry system of accounting, introduction of basic books of accounts of sole proprietary concern, control accounts for debtors and creditors, closing of books of accounts and preparation of trial balance, Final Accounts: Trading, profit and loss accounts and balance sheet of sole proprietary concern with normal closing entries, Introduction to manufacturing account, final accounts of partnership firms, limited company.

UNIT II

Introduction to computerized financial accounting: coding logic and codes required, master files, Transaction files, Introduction to documents used for data collection, processing of different files, outputs obtained,

UNIT III

Introduction to computerised Inventory control, types of inventory and associated documents, Inventory reports-nature and types, Inventory Control : ABC and Ageing analysis, Methods of Stock validation : LIFO, FIFO, actual bases, Interfacing Inventory with Financial Accounting, Purchasing Sub-Systems, Sales Order processing. Introduction to Computerised Payroll & Invoicing Applications, Exposure to: Structure, Processing and Reports, Interfacing these applications to financial Accounting.

UNIT IV

Use of Accounting package Tally: Introduction to Tally, Groups, Ledgers, Vouchers, Orders, Cost Centres and Categories. Stock. Reports in Tally

13 Suggested Resources for Reading

1. Kellock, J.: Elements of Accounting, Heinemann 1st ed., London.
2. Rockley, L.E.,: Finance for the Non-Accountant 2nd. Edition, London: Business Book.
3. Levy and Sarnat,: Principle of Financial Management, Prentice Hall International, PHI.
4. Arnold,: Financial Accounting, Prentice-Hall International (Paperback Edition).
5. Horngren and Sundem: Introduction to Financial Accounting, Prentice-Hall International (Paperback Ediion) N.D: PHI.
6. Murthy, U.S: Management Finance, 2nd. Edition, Vakils Fefers & Simons Ltd.
7. Van Home, James, C.: Financial Management & Policy, Prentice Inc.
8. Pandey, I.M.: Financial Management, Vikas Publication, 6th Rev. ed., N. Delhi.

Note: Latest edition of text books may be used.

SEMESTER V

SD501: Software Engineering and Quality Assurance

Objectives:

1. To make students familiar with all the software development principles, models and designing tools required to develop the software.

2. To make students understand theories, methods, and technologies applied for professional software development.

3. To discuss the concepts of software products and software processes.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.

2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

4. All questions carry equal marks.

Unit I

Introduction: Characteristics, Software Engineering goals, Software Process Models: Waterfall, Spiral, Prototyping, Fourth Generation Techniques and Software Development Life Cycle. Software Project Management: Software project management, Project planning and control, cost estimation, project scheduling using PERT and GANTT charts.

Unit II

Software Design: Basic issues in software design, modularity, cohesion, coupling and layering, function-oriented software design: DFD and Structure chart, Object-oriented software development, user interface design.

Unit III

Software Testing: Fundamentals of testing, White-box and black-box testing, Test coverage analysis and test case design techniques, Mutation testing, Static and dynamic analysis, Software Metrics, Types of Software Metrics.

Unit IV

Software Quality Assurance Concepts and Standards: Quality Concepts, Quality Control, Quality Assurance, SQA Activities, Software Reviews, Formal Technical Reviews, Review Guidelines, Software Reliability, Software Safety, Quality Assurance Standards, ISO 9000, ISO 9001:2000, Quality Factors, CMM, TQM, Six Sigma, SPICE, Software Quality Assurance Metrics.

References Resources:

1. Pressman, R.S., Software Engineering – A Practitioner's Approach, McGrawHill

2. Fairley, R.E., Software Engineering Concepts, McGraw Hill.

3. Jalota, P., An Integrated Approach to Software Engineering, Narosa PublishingHouse

4. Mall, R., Fundamentals of Software Engineering, PHI Learning Pvt. Ltd.

SD502:Computer Graphics and Multimedia Applications

Objectives:

1. To study the graphic techniques and algorithms and enable the students develop their creativity.

2. To write programs that demonstrates image processing techniques.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.

2. Examiner will set a total of NINE questions comprising two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

4. All questions carry equal marks.

Unit I

Introduction to Computer Graphics: Overview of Graphics Systems, Display Devices, Hardcopy Devices, Interactive Input Devices- Pointing and positioning devices (cursor,light pen, digitizing tablet, the mouse, track balls).

Unit II

Graphics Packages: Studying the Features and Developing Computer Graphics Using Standard Graphics packages like Auto CAD and Paint Brush.

Unit III

Applications of Multimedia and software packages: Multimedia in the Real World, Training and Education, Image Processing. Studying features and use of Multimedia Image Processing authoring tools like Photoshop and Macromedia Director.

Unit IV

Multimedia Technology: Multimedia in use: Introducing multimedia, uses of multimedia, Technology System Components, Multimedia Platforms, Development Tools, Image, Audio, Video, Storage for multimedia and Communications.

References Resources: :

1. Hearn, D. and Baker, M.P., Computer Graphics Principles and Practice, PHI.

2. Kanetkar, Y., Graphics Under 'C', BPB Publications.

3. Jeffcoate, J., Multimedia in Practice, Technology and Applications, PHI.

4. Hugles, J.F., Dam, A.V., McGuise, M., Splar, D.F., Foley, J.D., Feiner, S.K., & Akeley, K., Computer Graphics, Principles and Practice

5. Hughes, M., 'C'; Addison Wesley Publishers.

6. Sinclair, I. R., Multimedia on the PC (with CDROM), BPB Publications.

7. Hillman, D., Multimedia Technology and Applications, ITP.

8. Vaughan, T., Multimedia Making it Work, Osborne Publishers.

9. Kelly- Bootle, S., Turbo 'C', BPB Publications.

SD503:ASP.Net using C#

Objectives:

1. To make students familiar with development of console, windows and web applications using C# on .NET platform.

2. To improve object-oriented programming skills through practice and insights gained by studying a new programming language.

3. To gain a working knowledge of the C Sharp programming language and build object-oriented applications using C Sharp.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.

2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

4. All questions carry equal marks.

UNIT I

Introduction to ASP.NET: .NET Architecture, ASP.NET Fundamentals Compilation Modes of .Net Architecture, Understanding Client Server Architecture, Introduction to Web Technologies.

UNIT II

Standard Rich Controls: Templates in ASP. NET, Composite Data Bound Controls, Designing Website with Master Pages, Maintaining Application State, Microsoft Application Block, OOPS/COM/DLL/Assembly.

UNIT III

Globalization And Localization: Custom Controls, Web User Controls, Caching Application Pages and Data, Security, Java Scripts, JQUERY, AJAX .NET .

UNIT IV

Windows Mobile Application Development: Application Development mobile Browsers, Android Applications, LINQ, XML, XSLT, Web Services, WPF (SILVERLIGHT), WCF, Deployment.

Suggested Resources for Readings:

1. Jones, B. L., Sams Teach Yourself C# in 21 Days.

2. Balagurusamy, E., Programming in C#, Tata McGraw-Hill.

3. Liberty, J., Programming C#, O'Reilly.

4. Schildt, H., The Complete Reference: C#, Tata McGraw-Hill.

5. Robinson, S., Professional C#, Wrox Press.

6. Beres, J., SAMS Teach Yourself Visual Studio.Net in 21 days, Sams

SD504: Linux operating System

Objectives:

1. To get familiar with Linux system, its commands, files & directories, system, shell programming,

2. To introduce the student with Unix/Linux kernel programming techniques.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.

2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.

3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.

4. All questions carry equal marks.

UNIT I

Introduction to Linux : What is Linux, Linux's History, Minimum System Requirements; Installing Linux : Working with Linux, Floppy-less Installation, Boot and Root Disks, Choosing Text or Graphics Installation, Setting up your Hard Drive, Formatting the Partitions, Setting up the Ethernet, Configuration X, Selecting packages to Install, Using LILO; Partitioning the Hard Disk : Linux Swap Space Partitions, Linux's fdisk, Enabling the Swap Space for Installation, Creating the Linux File-system partition, Using LILO, GRUB

UNIT II

Using Linux : Starting and Stopping your Linux System, Linux Shutdown Commands, Login, Passwords, Creating a New Login, Logging Out; Trying out your new Login : Linux Error Messages, Search Paths; The who Command, Commands and Programs, Basic Linux Commands : How Linux Commands Work, Command Options, Other Parameters, Input and Output Redirection, National conventions used to Describe Linux commands, Online help available in Linux, The Linux Man pages, Finding keywords in Man pages, The bash shell help facility; Wildcards : * and ?, Environment Variables, Process and How to Terminate them, The process status Commands : ps, The process termination command : kill, the su command, the grep command.

UNIT III

Using the File System: Files Overview, Common types of files, filenames, Directories an Overview, Parent directories and sub-directories, The root directory, How directories are named, The home directory; File and Directory Permissions : File and Directoryownership, User and ownership, Groups, Changing group ownership, File Permissions, UMASK Setting, Changing File Permission, Changing directory permissions. Bash: What is Shell? How the Shell gets Started, The most common Shells; The Bourne Shell: Command-line Completion, Wildcards, Command History, Aliases, Input Redirection, Output Redirection, Pipelines Shell, Prompts, Job control, Customizing bash, bash commands, bash variables, tcsh : An Introduction to tcsh, Command completion, Wildcards, Command History, Aliases, Input and Output Redirection, Pipelines, Prompts, Job Control; Key Bindings, Correcting Spelling Errors, Pre-commands, Change directory Commands, Monitoring Logins and Logouts, Customizing tcsh, tcsh Command Summary, tcsh variables.

UNIT IV

Shell Programming: Creating and Running Shell Programs, Using variables: Assigning a value to a variable, Accessing the value of a variable, Positional Parameters and other Built-In Shell Variables; The Importance of Quotation Marks : The test Command, The tcsh Equivalent of the test command, Conditional Statements : if Statement , case Statement; Iteration Statements : for Statement, while Statement, until Statement, shift Command, select Statement, repeat

Statement, Functions. Editing and Typesetting : Text Editors vi, The vi Editor, Starting vi, vi modes, Inserting Text, Quitting vi, Moving the Cursor, Deleting Text, Copying and Moving Text, Searching and Replacing Text, Setting Preferences

Suggested Resources for Readings:

1. Parker, T., Linux Unleashed, Techmedia Publishing House.

2. Norton, P., Complete guide to LINUX, Techmedia.

3. Komarinski, M., LINUX System Administration Handbook, AW.

4. Goerzen, J., Linux Programming Bible, IDG Books

5. Venkateshmurthy, M.G., Introduction to Unix & Shell Programming, Pearson Education

SD505 Critical Thinking and Elementary Statistics

Objective: The objective of the paper is to provide knowledge to student about basics of statistics and to inculcate the habit of critical thinking.

Instructions for examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Introduction: Statistics as a Subject ; Statistical Data: Meaning and Types , Collection and Rounding of Data, Classification and Presentation of Data.

UNIT-II

Analysis of Univariate Data: Construction of a Frequency Distribution; Concept of Central Tendency, mean, median and mode.

UNIT-III

Dispersion and Their Measures; Time Series: Meaning, Components, Models, Fitting Linear and Quadratic Trend

UNIT-IV

Index Number: Meaning, Types, and Uses: Methods of Constructing Price and Quantity Indices (Simple and Aggregate).

Suggested Resources for Reading:

• Chou-YaLun: Statistical Analysis, Holt, Rinchart and Winslon.
• Levin, Richard I. & David S Rubin: Statistics for Management, Prentice Hall of India, Delhi.

References Resources:

1. Aczel, Amir D.: Complete Business Statistics, Irwin.

SD506 Introduction to Artificial Intelligence and Data Science

Objective: The course will help students to learn the basics of Artificial Intelligence and Data Science. It further aims to make them analyze data using spreadsheets.

Learning Outcomes

On successful completion of this course, a student will be able to

i. understand basics of Artificial Intelligence and Data Science.
ii. analyze data using spreadsheet software.
iii. visualize, search and summarize the data using spreadsheet.

Note :

i. The Question Paper will consist of Four Units.
ii. Examiner will set total of NINE questions comprising TWO questions from each Unit and ONE compulsory question of short answer type covering whole syllabi.
iii. The students are required to attempt ONE question from each Unit and the Compulsory question.
iv. All questions carry equal marks unless specified.

UNIT - I

Introduction to Artificial Intelligence (AI): Definition, Types of AI, History of AI, Applications of AI in healthcare, education and agriculture, Introduction to key technologies of AI: Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Robotics; Ethical issues and Future of AI. Problem Solving: AI problems, Agents and Environments, Structure of Agents.

UNIT - II

Data Science:Introduction to Data Science, evolution and need for Data Science, Types of Data, Data Science Applications in Various Fields, Data Science Life Cycle or Project Stages, Data collection, Data Pre-Processing Overview – Data Cleaning, Data Integration and Transformation, Data Reduction – Data Discretization, Data Storage and management, careers in Data Science

UNIT - III

Managing Datawith Spreadsheet Software:Creating a database, sorting& filtering data, custom sorting and advanced filtering, definingand grouping database ranges, applying data tools like data validation, removing duplicates, text to columns, Flash fill; Managing data using Math, statistical, Trigonometric, String/Text, Date and Time, logical and Database functions, Nested functions.

UNIT - IV

Data Visualization, Searching and Summarizing using Spreadsheet Software: Understanding Chart types, Creating Embedded Charts, formatting Charts: Changing Chart types, adding Titles, Legends and Gridlines, Printing Charts, searching data using HLOOKUP and VLOOKUP, summarizing data using pivot table, analyzing data using What-if Analysis: goal seek, scenario manager and data Table.

Suggested Readings:

i. D.Venu Gopal, Data Science and Artificial Intelligence for Beginners Paperback
ii. Sudhir Reddy Navar, Introduction to AI/ML and Data Science Paperback
iii. C. Raju, Data Science: A Beginner's Guide Paperback
iv. Sharma, A, Essentials of AI and Soft Computing, PHI Learning, 2024
v. Kevin Knight, Elaine Rich, B. Nair - Artificial Intelligence, TMH
vi. Manish Nigam, Data Analysis with Excel Paperback, BPB Publications
vii. Mali, L. ,. Libre Office 5.1 Writer, Calc: Math Formula Book- Vol 1. Notion Press.
viii. L. Winston Wayne, Microsoft Excel: Data Analysis & Business Model Paperback

SEMESTER VI

SD601: ENTREPRENEURSHIP DEVELOPMENT PROGRAMME

Objective: EDPs aim at training various target groups in entrepreneurial traits so that they obtain adequate information, motivation and guidance in setting up their own enterprises.

Instructions for examiner:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT-I

Entrepreneur: Definition, nature, and characteristics of entrepreneurship: Emergence of entrepreneurial class including women entrepreneurs: Theories of entrepreneurship: Socioeconomic environment and the entrepreneur. Characteristics of Entrepreneur Leadership, Risk taking, Decision making and business planning. Innovation and Entrepreneurship, entrepreneurial behaviour and motivation.

UNIT-II

Small Business as a Seed Bed of Entrepreneurship: Concept of business venture. The start-up process, Concept, Plan, Implementation, Initial Strategic Planning, Product and Marketing Scope, Legal and Tax consideration, Risk analysis and financial considerations. Profit Planning in Small Enterprise: Growth Strategies and diversification.

UNIT-III

Finance Management in current operations and expansion of capital. Role of Small Business in the national economy, National Policies for small business development, Governmental and Non- Governmental assistance. Contribution of Commercial Banks in promoting and servicing small business.Small business and modern technology.

UNIT-IV

Rules & Regulations: Governmental Policies and formalities in setting up a unit. Basic requirements regarding registration, Excise, Sales Tax, Factory Act, SSI exemptions: Licensing and Registration procedure; Appreciation of important provisions of Factory Act, Shops & Commercial Establishment Act; Sales of Goods Act, Partnership Act; Contract Act; Income Tax, Sales Tax and Excise rules; Insurance.

Suggested Resources for Reading:

1. 'Project Engineering & Management' by A.K. Sinha, Vikas Publishing House Pvt. Ltd., 1983.
2. 'Project Planning, Financing, Implementation & Evaluation' by U. K. Srivastava, Indian Institute of Management, Ahmedabad, 1981.
3. 'Financial Management - An Analytical and Conceptual Approach' by Kuchhal, S. C., Chaitanya Pub. House, 1982.
4. 'Principles of Management Accounting' by Mohan &Goyal, Agra SahityaBhavan, 1982.
5. 'Management of Small Scale Industries' by Saroja, Seth Publishers, Bombay, 1979.

References Resources:

1. 'How to Succeed in Small Industry' by Vepa Ram K., Vikas Publishing House, New Delhi, 1984.
2. Bare Acts: Central Sales Tax Act, State Sales Tax Act, Central Excise Act and Customs Act.
3. 'Law and Practice of Income Tax in India' by Bhagwati Prasad, NavmanPrakashan, 1972.
4. 'A Text Book of Commercial Law' by S. S. Gulshan, S. Chand & Co., 1979
5. 'Industrial Relations' by B.P.Gupta, PHD Chamber of Commerce &Inds, 1986

SD 602:E-Commerce

Objectives:

1. To integrate the knowledge of foundational functional areas of commerce in order to develop a holistic perspective on the role of IT in organizations.
2. To discuss emerging e-commerce topics.
3. To apply the concepts of Internet security and multimedia in e-business applications.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

Unit I

Introduction - E-commerce. Advantages, Types and Classification of e-commerce, Building the Web site, Internet, Online transaction, Shopping Card, Software, Risk of e- commerce, email Ids, Introduction to Cookies and Sessions Digital Signature: Digital Signature and its Application in E-Commerce, E-Commerce Privacy and Trust.

Unit II

Enterprise Resource Planning: The Emergence of ERP Systems, Business benefits of ERP, ERP Modules and design alternatives, challenges of ERP implementation, Business Processed engineering, ERP system development process.

Unit III

Mechanism of making payment through internet: Online payment mechanism; Electronic Payment systems, payment Gateways, Visitors to website, tools for promoting websites; Plastic Money: Debit Card, Credit Card.

Unit IV

Growth of E-Commerce - Present and potential Data and Measurement issue for E- Commerce. The infrastructure for E-Commerce, Recent and Near terms, Video Conferencing and Internet. Entertainment Education Health Services, Professional Services, Publishing, Financial Services, Web Site Evaluation.

Suggested Resources for Readings:

1. KalaKota, R. & Whinston, A. B., Frontiers of Electronic Commerce; Addison Wesley.
2. Sumner, M., Enterprise Resource Planning, Pearson Education.
3. Bajaj, K. & Nag, D., E-Commerce : The Cutting Edge of Business; TataMcGraw-Hill.
4. Ray, R., Enterprise Resource Planning : Text and Cases, TMH.
5. Minoli, D. & Minoli, E., Web Commerce Technology Handbook, Tata McGraw Hill.

SD603: Emerging Technologies in Computing

Objectives:

1. This course enables students to be familiar with Emerging Technologies such as Parallel Computing, Cloud Computing, Green Computing and Data Mining & data warehousing.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

Unit-I

Parallel Computing: Concept of Parallel Computing, Flynn's Classical Taxonomy: SISD, SIMD, MISD, MIMD; Parallel Computer Memory Architectures: Shared Memory, Distributed Memory, Hybrid Distributed-Shared Memory; TDB and BNP Class of Scheduling Algorithms.

Unit-II

Cloud Computing: Introduction and use, Architecture, Service Models: infrastructure as a service, platform as a service, and software as a service, Introduction to IoT

Unit-III

Green Computing IT Fundamentals: Business, IT, and the Environment. Green Assets: Buildings, Data Centers, Networks, and Devices

Unit-IV

Data mining & Warehouse: KDD versus data mining, Stages of the Data Mining Process- task primitives, Data Mining Techniques -Data mining knowledge representation – Data mining query languages, Integration of a Data Mining System with a Data Warehouse

Suggested Resources for Readings:

1. Han, J. and Kamber, M. & Dei, J., Data Mining: Concepts and Techniques, Morgan Kaufmann Publishers
2. Berson, A., and Smith, S.J., Data Warehousing, Data Mining & OLAP, Tata McGraw Hill.
3. Gupta, G. K., Introduction to Data Min Data Mining with Case Studies, Easter Economy Edition, Prentice Hall of India,
4. Joseph, J. & Fellenstein, C., Grid Computing, Pearson Education.
5. Sasikumar, M., Shikhare, D., & Prakash, P.R., Introduction to Parallel Processing, PHI.
6. Lamb, J., The Greening of IT, Pearson Education.
7. Harris, J., Green Computing and Green IT- Best Practices on Regulations & Industry, Emero.

SD604: PHP Programming

Objectives:-This course will enable the students to learn the PHP language for website Development.

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

UNIT I

Introducing PHP – Basic development Concepts, Web Server Definition and concept, Web Application Definition – Creating first PHP Scripts – Using Variable and Operators – Storing Data in variable – Understanding Data types – Setting and Checking variables Data types – Using Constants – Manipulating Variables with Operators.

UNIT II

Controlling Program Flow: Writing Simple Conditional Statements - Writing More Complex Conditional Statements – Repeating Action with Loops – Working with String and Numeric Functions.

UNIT III

Working with Arrays: Storing Data in Arrays – Processing Arrays with Loops and Iterations, Types of Arrays, Super Global Variables, Using Arrays with Forms – Working with Array Functions – Working with Dates and Times. Creating Forms

UNIT IV

Using Functions: Creating User-Defined Functions, Introduction to PHPMYADMIN Working with Database and SQL: Using MYSQLi, Different mysqli functions to create a database, creating tables, fetching records from database, update table etc. Introduction to session and cookie. Working with Files and Directories: Reading Files- Writing Files-Processing Directories.

Suggested Resources for Readings:

1. Holzner, S., PHP: The Complete Reference, McGraw Hill Education
2. Gupta, A., PHP: The Ultimate Beginners Guide to PHP for Beginners, Independently Published.

SD605: Network Management

Objectives:-

1. This course will enable the students to manage the network devices and configuring servers

Instructions:

1. The syllabus of this paper has been divided into FOUR units.
2. Examiner will set a total of NINE questions comprising Two questions from each unit, including Question No. 1 (compulsory) of short answer type covering the whole syllabus.
3. The students are required to attempt one question from each unit and the entire Compulsory Question No. 1.
4. All questions carry equal marks.

Unit-I

Management of Network Devices:- Managing network devices :- Routers, Switches, hubs Startup and Configuration

Unit-II

Configuring and installing Window based servers:- Configuring IP addresses, Installing and Configuring Windows Server - Preparing for Installation, Creating windows server boot disk, Installing windows server, Configuring server/ client

Unit- III

Configuring & installing Linux based Servers:- Configuring IP addresses , steps for installing linux based server, installation of drivers , Configuring server on the network

Unit-IV

Windows Networking: Sharing files and resources: - User Accounts, Groups, Permissions & Their Role in Sharing, The Basic Concepts in Network Sharing, Customizing Your Network Sharing Settings, How to Share Devices with Others on the Network.

References Resources: -

1. Stuart, E., Network Management: Concepts and Tools, Sprinder
2. Zacker, C., Installing & Configuring Windows Server 2012 R2, Microsoft Press
3. Wilson, G., Open Linux Installation & Configuring Handbook

SD606: Computer Architecture

Objectives: This course will enable the student to understand the basic computer organization and its internal hardware architecture design.

Learning Outcomes :

On successful completion of the course, students will be able to:

i. Understand basic computer organization and its architecture.
ii. Learn how micro-operations, interrupt and instruction cycles are executed in logic circuits.
iii. Learn basic features, memory, input output organization of a simple microprocessor.

Note :

i. The Question Paper will consist of Four Units.
ii. Examiner will set total of NINE questions comprising TWO questions from each Unit and
iii. ONE compulsory question of short answer type covering whole syllabi.
iv. The students are required to attempt ONE question from each Unit and the Compulsory question.
v. All questions carry equal marks unless specified.

UNIT I

Definition of computer organization, design and computer architecture ; Digital Systems : basic block diagram of computer ; ALU design : Register Transfer Language, bus and memory transfer; Microoperations and their hardware implementation: Arithmetic microoperations - binary adder subtractor, binary incrementer, arithmetic circuit; Logic microoperations – logic circuit; Shift microoperations – 4 bit combinational shifter; Arithmetic Logic Shift Unit.

UNIT II

Basic Computer Organization : stored program organization, Von Neumann architecture ; microoperations and macrooperations ; instruction code format, direct and indirect addressing ; basic computer registers ; Types of instructions - memory reference, register reference , input output instructions ; Common bus system; instruction cycle ; interrupt cycle ; types of interrupts ; Introduction to assembly language , assembly language vs machine language.

UNIT III

Memory organization : memory hierarchy ; RAM and ROM chips, memory connection of four 128x8 RAM and one 512x8 ROM chip to CPU ; Associative memory; Cache memory - associative, set associative and direct mapping; Virtual memory - paging and segmentation; Microprocessor architecture: 8086/8088 - features, block diagram, memory and register organization, flag register, addressing modes.

UNIT IV

Input Output organization : input output interface ; IOP design, isolated I/O , memory mapped I/O; Asynchronous data transfer: source initiated - destination initiated strobe control and handshaking modes of transfer ; programmed I/O data transfer, interrupt initiated data transfer ; Direct Memory Access : DMA controller, cycle stealing and burst mode DMA transfer.

Suggested Readings:

1. Mano, M. Morris: Computer System Architecture, Third Edition, Pearson Education India, 2007.
2. Mano, M. Morris, Contributer Mall, Rajib : Computer System Architecture, Revised Third Edition, Pearson Education India, 2017.
3. Stallings, William : Computer Organization and Architecture : Designing for Performance, 11th Ed. Pearson Education India, 2022.
4. Hayes , John P. : Computer Architecture and Organization, 3rd Ed, McGraw Hill Education, 2017.
5. Chaudhuri, P.Pal : Computer Organization and Design, 3rd Ed, Prentice Hall of India Learning Pvt. Ltd., 2008.

SD 607: Introduction to Machine Learning

Objective: The course aims at introducing basic concepts of machine learning and implementation of machine learning algorithms.

Learning Outcomes:

On successful completion of the course, students will be able to:

i. Learning the basics of Machine Learning and its applications.
ii. Understanding of data preprocessing, regression use in machine learning and data reduction.
iii. Understanding of supervised learning algorithms and its implementation.
iv. Understanding of unsupervised learning algorithms and its implementation. Basic understanding of deep learning.

Note :

i. The Question Paper will consist of Four Units.
ii. Examiner will set total of NINE questions comprising TWO questions from each Unit and ONE compulsory question of short answer type covering whole syllabi.
iii. The students are required to attempt ONE question from each Unit and the Compulsory question.
iv. All questions carry equal marks unless specified.

UNIT-I

Introduction: Concept of Machine Learning, Supervised and Unsupervised learning, Training vs Test Data, Reinforcement learning, Designing a Learning System, Issues in Machine Learning, Applications of Machine Learning.

UNIT-II

Data Preprocessing: Understanding of data and its preprocessing, normalizing data, feature scaling and feature selection techniques, Overfitting, Data reduction using Principal Component Analysis.

UNIT-III

Supervised Learning: Concept of Classification, Decision tree, k-nearest neighbor, Naïve Bayes Classifier, Support Vector Machine, Neural Networks and backpropagation algorithm, Classification evaluation metrics.

UNIT-IV

Unsupervised Learning and Deep Learning: K-means Clustering, Limits of K-Means, DBSCAN, Concept of Deep Learning, Architecture of Convolutional Neural Networks and Recurrent Neural Networks.

Suggested Readings:

1. Andreas . C. Müller and S. Guido, "Introduction to Machine Learning with Python," O'Reilly, 2017
2. Amanda . Casari and Alice . Zheng, "Feature Engineering for Machine Learning," O'Reilly Media, Inc., 2018,
3. Sharma, A., Essentials of AI and Soft Computing, PHI Learning, 2024
4. Mitchell, T.M., Machine Learning, McGraw Hill Education, 2017.
5. Ian Goodfellow, Yoshua. Bengio, and Aaron. Courville, "Deep Learning," MIT Press, 2016
6. Alpaydin, E., Introduction to Machine Learning, MIT press, Fourth edition, 2020.